Calibrating Minimum RTT Under Low ACK Frequency

draft-li-quic-minimum-rtt-estimation-01

Tong Li, Ke Xu, Bo Wu, Youjian Zhao

IETF-125, QUIC WG, March, 2026

Overview of Standards on ACK Mechanism

Reducing ACK frequency is standardizing in QUIC WG

Legacy TCP RFCs

Delayed ACK Baseline

- RFC 1122 / 5681
- RFC 4341 / 5690 / 3449

fixed ACK frequency

RFC 9000

QUIC Default ACK

Delayed ACK by default and immediately on reordering

baseline QUIC behavior

SIGCOMM 2020

TACK [1]

ACK frequency can be determined by the connection's bandwidth-delay product (BDP)

research milestone

IETF QUIC WG

IYENGAR-ACK [2]

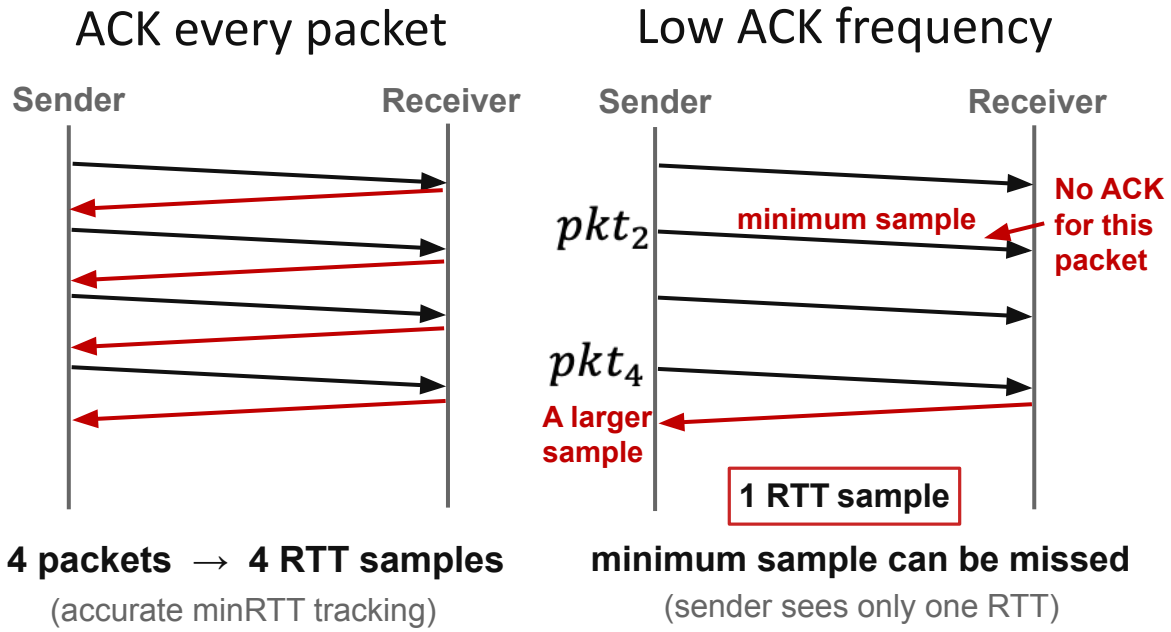
Adds a QUIC extension for negotiating lower ACK frequency

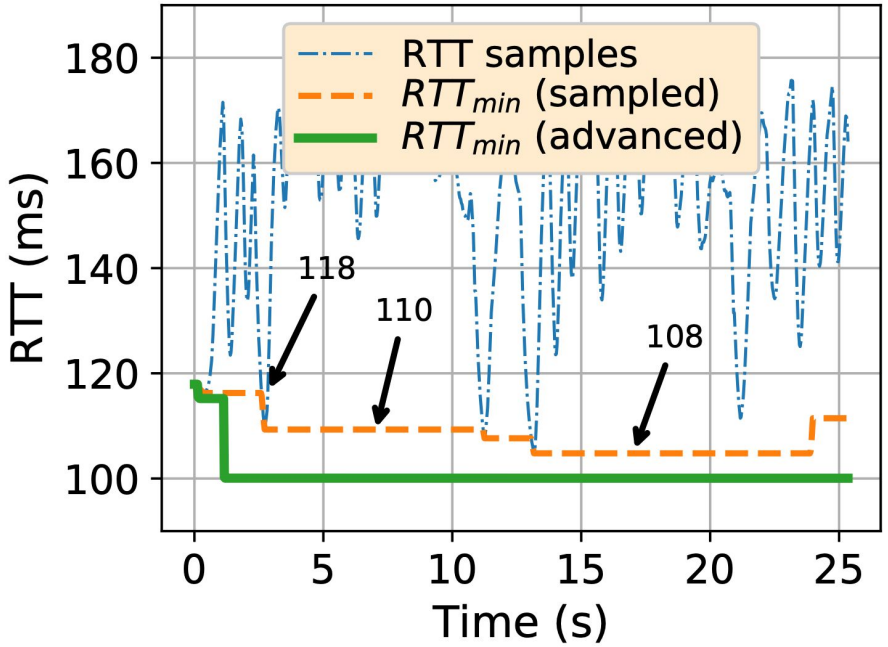
standardization path

[1] TACK: Improving Wireless Transport Performance by Taming Acknowledgments. ACM SIGCOMM, 2020.
[2] Jana Iyengar, Ian Swett, Mirja Kühlewind. "QUIC Acknowledgment Frequency". Work in Progress Internet-Draft, draft-ietf-quic-ack-frequency-14, February 2026

Issues Under Low ACK Frequency

minRTT: minimum RTT

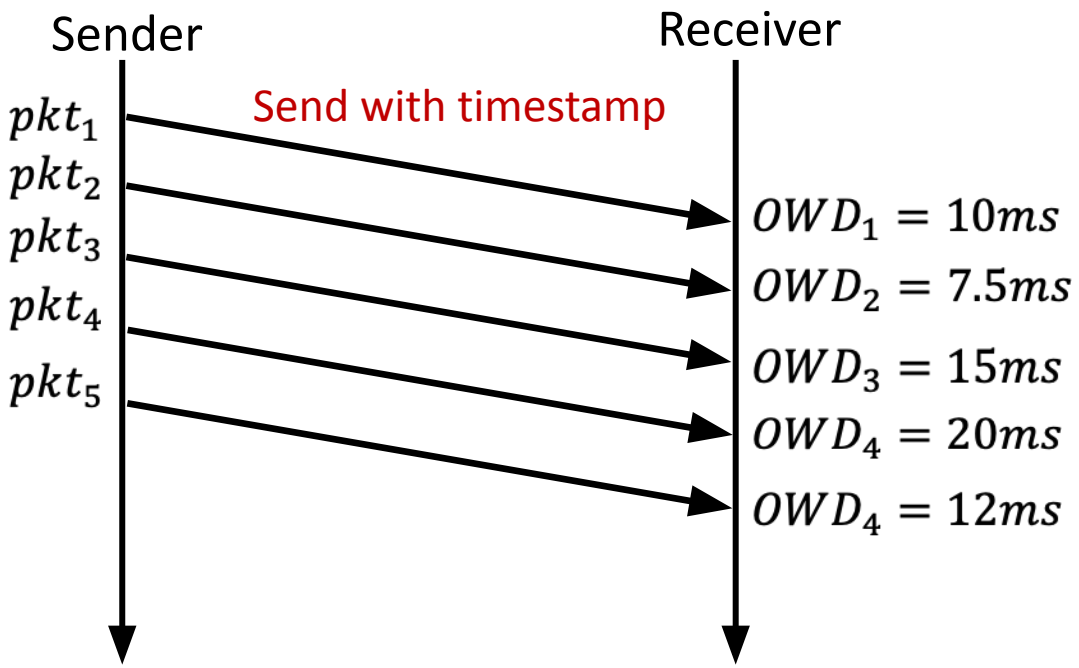




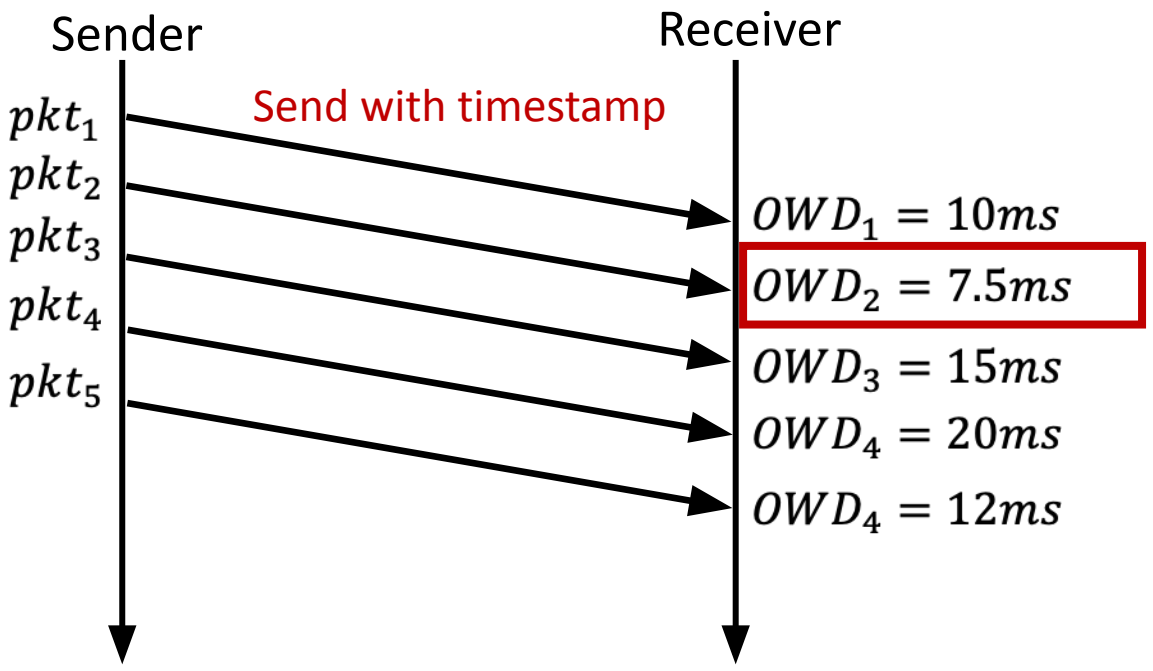
- Fewer ACKs ⇒ One RTT sample for many packets ⇒ Biased MinRTT
- Experiments in Wi-Fi links report **8%–18%** larger minRTT estimates
- *Worse in satellite links...*

How to calibrate MinRTT?

One-Way Delay (OWD) to the rescue!



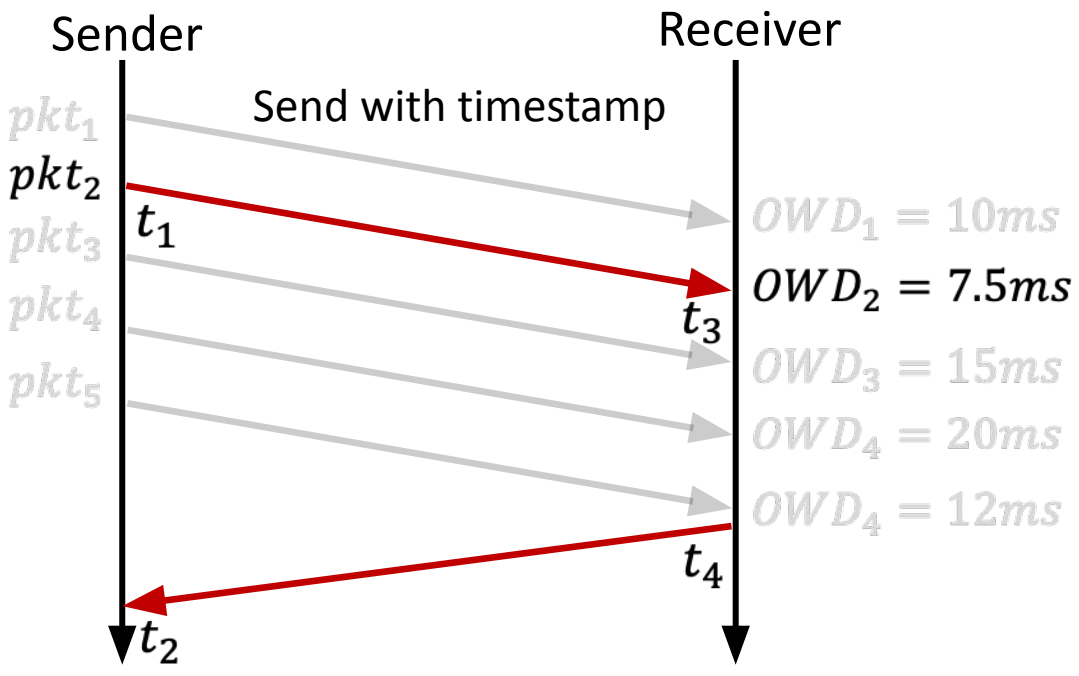
MinRTT Estimation Under Low ACK Frequency



minOWD
measured at the receiver

minOWD: minimum OWD

MinRTT Estimation Under Low ACK Frequency



minOWD: minimum OWD

$$minRTT_{sample} = t_2 - t_1 - ACK\ Delay$$

$$ACK\ Delay = t_4 - t_3$$

- t_1: Departure time of packet that achieves minOWD
- t_2: Arrival time of ACK frame
- t_3: Arrival time of packet that achieves minOWD
- t_4: Departure time of ACK frame

What’s new for QUIC protocol

1. **Transport Parameter: timestamp-support**

A new field named timestamp-support should be added for negotiation between both parties on whether to sync packet departure timestamps in QUIC connection.

2. **TIMESTAMP Frame**

A new frame is needed to exchange per-packet departure timestamps, and its format can reuse the TIMESTAMP frame proposed in an existing QUIC draft [3].

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                            0x XX (i)                        ...
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Departure Timestamp (i)                ...
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

3. **MINOWD-ACK Frame**

A new frame identifies the packet with the minimum one-way delay within an ACK Delay and Departure Timestamp.

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                            0x XX (i)                        ...
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      MinOWD ACK Delay (i)                   ...
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      MinOWD Departure Timestamp (i)         ...
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

[3] Christian Huitema, QUIC Timestamps for Measuring One-Way Delays, Work in Progress, Internet-Draft, draft-huitema-quic-ts-08, August 2022. Progress Internet-Draft, draft-huitema-quic-ts-08

Conclusion

1. Low ACK frequency biases minRTT Estimation

2. New method to calibrate minRTT estimation

- Receiver calculates per-packet OWD, reports timestamp selectively
- Sender calculates RTT samples according to timestamps

3. QUIC Modification

- Transport parameter: Timestamp-support
- Recall draft-huitema-quic-ts-08: TIMESTAMP frame
- Add a new MINOWD-ACK frame: Report ACK Delay and Departure TS of pkt_minOWD

Any comments welcome

FAQs

Q1: Why minRTT estimation **important**?

A2: minRTT is a fundamental state variable in transmission control

- Used by BBR to estimate congestion window [CARDWELL-BBR]
- Used by QUIC loss detection to reject implausibly small RTT samples [RFC9002]
- Used to update the ACK frequency in [SIGCOMM 20]

Q2: Why use **OWD** to calibrate minRTT?

A2: OWD change reflects RTT change in most cases. OWD can deliberately identify the packet with the minRTT instead of hoping legacy ACK sampling happens to catch it.

Q3: Need clock synchronization for one-way delay?

A3: No, it only relies on **comparing relative OWDs** using the receiver's local clock and reporting the packet that achieves the minimum OWD.

Q4: Why receiver report **departure timestamp** instead of **packet number** of pkt_ minOWD?

A4: Sender can compute minRTT sample according to departure timestamp or packet number of the packet that achieves minOWD. However, departure timestamp feedback **avoids sender-side table lookup.**